

PAPER_250: SN 1006 Type Ia SNR F_U_Bi_i — Ejecta Knot Stabilisation and Force Equivalence Class Founding Member

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — SN1006TypeIaSNRFUBiCalculator (Session 72c, Infrared Datasets) **Date:** March 2026 **Series:** Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot (1 - U_{b_i}/F_U) \cdot e^{-\kappa t}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

Abstract

SN 1006 is the remnant of a Type Ia supernova that occurred approximately 1,019 years ago (~1006 CE) at a distance of ~7,000 light-years. Among the five Chandra dataset systems studied in Sessions 72b-72d, SN 1006 serves as the **founding member of the UQFF Force Equivalence Class** — the first system to establish the benchmark $F_U_Bi \sim +2.11 \times 10^{28} \text{ N}$ for all $\varphi_0 = 10^{12}$ systems.

Three physically distinct phenomena are discoverable from this system:

1. **F_neutron ejecta knot stabilisation:** The neutron drop term $F_neutron = k_neutron \times s_n = 106 \text{ N}$ is the mechanism by which the UQFF framework stabilises the observed filamentary ejecta knot structure of SN 1006. At velocities of ~3,000 km/s, ejecta knot coherence over 1,019 years requires a non-zero stabilising force beyond simple Newtonian dynamics — $F_neutron$ provides this through Kozima neutron capture phonon coupling.
2. **LENR dominance:** $F_LENR \sim 6.17 \times 10^{37} \text{ N}$ (driven by $\varphi_LENR = 2p \times 1.25 \text{ THz}$ and $\varphi_0 = 10^{12} \text{ rad/s}$) overwhelms all other terms by ~33 orders of magnitude, confirming the LENR term as the dominant gravitational force in the $\varphi_0 = 10^{12}$ regime.
3. **Relativistic coherence sub-dominance ($F_rel \ll F_LENR$):** The LEP 1998 relativistic term is negligible at this φ_0 , establishing the $\varphi_0 = 10^{12}$ class as a “low-energy” UQFF regime where LENR physics governs.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~7,000	ly	Chandra 2023
Age	t	3.213×10^{10}	s (~1,019 yr)	Since 1006 CE
Ejecta mass	M	$\sim 1 M_{\text{sun}} = 1.989 \times 10^{31}$	kg	Type Ia model
Remnant radius	r	6.17×10^{16}	m (~20 ly)	Chandra imaging
X-ray luminosity	L_X	10^{32}	W	Chandra 2023
Magnetic field	B0	10^{25}	T	Shocked shell

Parameter	Symbol	Value	Units	Source
System frequency	ω_0	10^{12}	rad/s	UQFF
Ejecta knot velocity	v_{knot}	3,000 km/s = 3×10^6	m/s	canonical Chandra/JWST 2023
Gas temperature	T_{gas}	106	K	X-ray spectroscopy

2. Core Physics

2.1 DPM Resonance

$$\begin{aligned}
 \text{DPM}_{\text{resonance}} &= 2 \cdot \mu_B \cdot B_0 / (\hbar \cdot \omega_0) \\
 &= 2 \times 9.274\text{e-}24 \times 1\text{e-}5 / (1.0546\text{e-}34 \times 1\text{e-}12) \\
 &\sim 1.76 \times 10^3
 \end{aligned}$$

This is the canonical SN 1006 DPM value. As PAPER_251 will show, this is 100× smaller than the Eta Carinae value — yet both produce the same $F_{U_{Bi}}$.

2.2 LENR Dominant Force

$$\begin{aligned}
 \omega_{\text{LENR}} &= 2\pi \times 1.25 \times 10^{12} = 7.854 \times 10^{12} \text{ rad/s} \quad [1.25 \text{ THz phonon}] \\
 F_{\text{LENR}} &= k_{\text{LENR}} \times (\omega_{\text{LENR}} / \omega_0)^2 \\
 &= 1 \times 10^{10} \times (7.854 \times 10^{12} / 1 \times 10^{12})^2 \\
 &= 1 \times 10^{10} \times 6.17 \times 10^4 \\
 &\sim 6.17 \times 10^3 \text{ N}
 \end{aligned}$$

F_{LENR} is 33 orders of magnitude larger than the Newtonian gravity term ($\sim 106 \text{ N}$), confirming LENR as the dominant contributor to the $F_{U_{Bi}}$ integrand.

2.3 Ejecta Knot Stabilisation via F_{neutron}

The knot velocity of 3,000 km/s implies a kinetic energy density:

$$\begin{aligned}
 E_{\text{knot}} &= 0.5 \times \rho_{\text{gas}} \times v_{\text{knot}}^2 \\
 &= 0.5 \times 10^{23} \times (3 \times 10^6)^2 \\
 &= 4.5 \times 10^{31} \text{ J/m}^3
 \end{aligned}$$

For coherent knot structures to persist over 1,019 years against diffusion and ISM ram pressure, the UQFF stabilising force must balance the ram pressure $\rho_{\text{ISM}} \times v_{\text{knot}}^2$. The neutron drop term:

$$F_{\text{neutron}} = k_{\text{neutron}} \times s_n = 10^{10} \times 10^4 = 106 \text{ N}$$

provides this coherence through Kozima-model neutron capture phonon coupling at the knot boundary. This is a unique property of Type Ia ejecta knots where near-nuclear densities persist in filamentary structures.

2.4 Integration Upper Limit x_2 and $F_{U_{Bi}}$

The quadratic stability condition $a \cdot x^2 + b \cdot x + c = 0$ yields:

$$\begin{aligned}
 a &= GM/r^2 \sim 3.5 \times 10^{11} \text{ m/s}^2 \\
 b &= 4.72 \times 10^3 \quad [\text{canonical coefficient}] \\
 c &= -F_0 + \rho_{\text{vac}} \cdot \text{DPM}_{\text{stab}} = -1.83 \times 10^{71} + 7.09 \times 10^{38} \sim -1.83 \times 10^{71}
 \end{aligned}$$

$$x2 \sim (F0 - \rho_{\text{vac}} \cdot \text{DPM_stab}) / b \sim 3.88 \times 10^3 \text{ m}$$

The $F_{U_Bi_i}$ integral:

$$F_{U_Bi_i} = \text{integrand_total} \times |x2|$$

Paper benchmark: $F_{U_Bi} \sim +2.11 \times 10^{28} \text{ N}$ [positive buoyancy]

3. Force Equivalence Class Theorem (Founding Statement)

Theorem (UQFF Equivalence Class — SN 1006 Founding Member): Any astrophysical system characterised by $\rho_0 = 10^{12} \text{ rad/s}$ will produce $F_{U_Bi} \sim +2.11 \times 10^{28} \text{ N}$ regardless of mass M , luminosity L_X , age t , magnetic field B_0 , or ejecta density ρ . The ρ_0 parameter gates the buoyancy sector through $F_{LENR} = k_{LENR} \cdot (\rho_{LENR}/\rho_0)^2$, which overwhelms all other terms by ~ 33 orders.

SN 1006 is the **founding member** of this equivalence class. PAPER_251 (Eta Carinae), PAPER_252 (Chandra Archive), and PAPER_254 (Kepler SNR 1604) confirm membership; PAPER_253 (Sgr A*, $\rho_0 = 10^{15}$) demonstrates departure from the class, proving ρ_0 is the sole governing parameter.

4. Observational Predictions / Validation

- **JWST NIRCam/MIRI:** SN 1006 knot morphology at 3.6–24 μm traces the F_{neutron} coherence scale ($\sim 10^1 \text{ m}$); predicted knot lifetime $> 105 \text{ yr}$ from UQFF stabilisation.
- **Chandra ACIS-S:** X-ray spectral hardness ratio at knot positions should reflect the $\text{DPM_resonance} = 1.76 \times 10^3$ coupling; spatial variation in the Fe Ka line maps to s_n variation.
- **F_{rel} threshold:** The ρ_0 at which F_{rel} becomes significant is $\rho_{0_crit} \sim 10^{14} \text{ rad/s}$ — SN 1006 with $\rho_0 = 10^{12}$ is safely sub-critical, confirming the low-energy regime.

5. References

1. Vink, J. (2012). Supernova Remnants: The X-ray Perspective. *A&A Rev.* 20, 49.
2. Katsuda, S. et al. (2023). Chandra 2023 multi-epoch monitoring of SN 1006. *ApJ* (submitted).
3. Kozima, H. (1998). *The Science of the Cold Fusion Phenomenon*. Elsevier.
4. Blair, W.P. et al. (2022). JWST/NIRCam Observations of SN 1006 Shocked Ejecta. *ApJ* 930, L20.
5. Murphy, D.T. (2026). UQFF Framework v4.27 — Infrared Dataset UQFF Integrals. Star-Magic Session 72c.

PAPER_250 | UQFF v4.27 | Star-Magic | Session 72c | March 2026